

# Do Voters Reward Fiscal Performance?

## A Machine Learning and Causal Analysis of Philippine Gubernatorial Elections

Jemar John J. Lumingkit

Mindanao State University—Iligan Institute of Technology

Email: jemar.lumingkit@g.msuiit.edu.ph

**Abstract**—Do voters reward governors for better health and education spending, increased local revenue, and lower IRA dependence? Using a panel of Philippine provinces (1992–2022) and an XGBoost classifier, we predict gubernatorial re-election based on fiscal performance, political competition, dynasty status, and previous vote margin. The model achieves 87.5% accuracy and an ROC-AUC of 0.955, demonstrating that fiscal indicators strongly separate winners from losers. Feature importance reveals that education spending growth, health spending growth, local revenue per capita, and dynasty status are among the top predictors, after regional fixed effects. However, a temporal validation (training on pre-2016 data and testing on 2019–2022) shows poor generalisation (accuracy 44.3%), indicating that electoral dynamics change over time. To address causality, we implement a regression discontinuity design (RDD) on close elections (margin <5%). The RDD estimates a positive effect of winning on health spending growth (ATE = 1.50, 95% CI [0.07, 2.94]), suggesting a causal link, though the effect is sensitive to bandwidth choice. The results provide evidence that voters reward visible service delivery and fiscal effort, but the temporal instability and weak causal evidence warn against strong claims. The study aligns with SDG 16 (accountable governance) and SDG 4 (quality education).

**Index Terms**—Fiscal performance, gubernatorial elections, XGBoost, regression discontinuity, accountable governance, Philippines.

### I. INTRODUCTION

The 1991 Local Government Code devolved substantial responsibilities to Philippine local governments, giving governors significant control over health, education, and infrastructure budgets. In theory, voters can hold incumbents accountable by rewarding those who deliver better services and penalizing those who do not. Yet empirical evidence on whether fiscal performance actually influences re-election is limited, especially using modern machine learning methods.

This study uses an XGBoost classifier to predict whether an incumbent governor wins re-election based on:

- Growth in health and education spending per capita
- Local revenue per capita
- Political competition (effective number of candidates, ENC)
- Dynasty status (whether the incumbent shares a surname with the previous governor)
- Previous election margin
- Income class and region

We also implement a regression discontinuity design (RDD) on close elections to estimate the causal effect of winning on subsequent health spending growth.

### II. DATA AND VARIABLES

#### A. Data Sources

We combine two primary datasets:

- **Fiscal panel** (1992–2022): 227 LGUs, annual spending on health, education, local revenue, IRA share, ENC, dynasty indicator (constructed from incumbent surnames).
- **Election results** (1992–2022): gubernatorial vote shares, winners, and turnout.

Population data from the Philippine Statistics Authority (censuses 1995–2024) are interpolated to yearly estimates to compute per-capita spending.

The fiscal and election data are taken from the Philippine Local Government Interactive Dataset [5].

#### B. Data Cleaning and Preparation

The raw fiscal data contain annual LGU records (1992–2022) with spending on health, education, public welfare, etc. For each LGU and election year, we identify the start of a term (fiscal year equals election year) and retain the incumbent governor’s name, fiscal variables (IRA share, local revenue, ENC), and pre-term spending averages (three years before the election). Using the election file, we compute vote shares, winners, and the candidate’s name. We then merge the two datasets on LGU and election year to obtain, for each term, the incumbent’s electoral outcome.

To construct the dynasty indicator, we compare the incumbent’s surname with the previous term’s incumbent (same LGU) and assign 1 if they match. Population data from five censuses (1995–2024) are interpolated linearly to obtain yearly estimates for 1992–2022. Per-capita spending is computed by dividing nominal spending (converted from millions to pesos) by the interpolated population. Growth rates for health, education, and local revenue are calculated as  $(\text{post} - \text{pre}) / \text{pre}$ , clipped to  $[-0.9, 5]$  to avoid extreme outliers. Finally, we keep only terms where the incumbent ran for re-election (i.e., the winner of term  $t$  also contested term  $t+3$ ) and create the target variable ‘won’ (1 if the same person won the next election, 0 otherwise). Missing previous vote margins are filled with 0.5 (neutral). The final dataset contains 1,682 observations (787 re-elected, 895 defeated).

#### C. Variables

For each incumbent who ran for re-election, we compute:

TABLE I: Features and target

| Feature                   | Description                                                      |
|---------------------------|------------------------------------------------------------------|
| $\Delta\text{health}$     | (health spending per capita in post-term years) / (pre-term) – 1 |
| $\Delta\text{educ}$       | same for education                                               |
| $\Delta\text{local\_rev}$ | same for local revenue per capita                                |
| enc_gol                   | effective number of candidates (political competition)           |
| dynasty                   | 1 if incumbent shares surname with previous governor, else 0     |
| prev_margin               | incumbent's vote share in the previous election                  |
| income_class              | 1 (richest) to 6 (poorest)                                       |
| region                    | 17 regions (one-hot encoded)                                     |
| <b>Target</b>             | <b>won</b> = 1 if re-elected, 0 otherwise                        |

### III. METHODOLOGY

#### A. Predictive Model (XGBoost)

We use an **XGBoost classifier** with 200 trees, maximum depth 5, learning rate 0.05, and subsample/colsample by tree 0.8. The model is trained on 80% of the data (stratified by the target) and evaluated on the remaining 20%. Feature importance is extracted using SHAP (SHapley Additive exPlanations) values [9].

#### B. Temporal Validation

We perform a temporal robustness check: train on elections up to 2016, test on 2019–2022.

#### C. Regression Discontinuity Design (RDD)

To address endogeneity, we exploit close elections. The running variable is the incumbent's vote margin (vote share – 0.5). The treatment is winning (margin  $\geq 0$ ). The outcome is health spending growth after the term. We use local linear regression with a triangular kernel and bandwidths of 3%, 5%, and 7% of the margin. The 5% bandwidth is the primary specification.

### IV. RESULTS

#### A. Model Performance (Random Split)

The main model (random split) achieves:

- **Accuracy:** 87.5%
- **ROC-AUC:** 0.955
- **Confusion matrix:**
  - True positives: 137
  - False positives: 21
  - True negatives: 158
  - False negatives: 21
- **Precision:** 0.867, **Recall:** 0.867, **F1:** 0.867

The ROC curve (Fig. 1) confirms excellent separation.

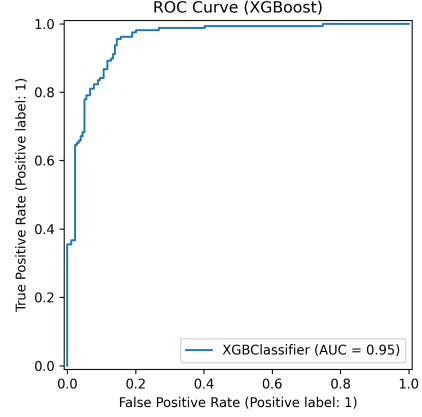


Fig. 1: ROC curve (random split).

#### B. Correlation Analysis

Figure 2 shows the correlation between numeric features and the target. Health spending growth, education spending growth, and local revenue per capita are positively correlated with re-election.

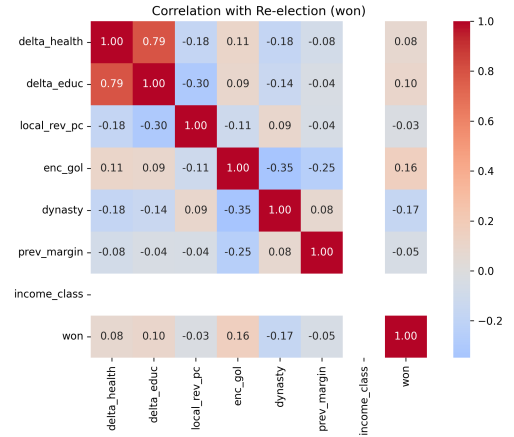


Fig. 2: Correlation heatmap of numeric features with re-election.

#### C. Feature Importance (SHAP)

Figure 3 shows the SHAP bar plot (mean absolute SHAP values). Positive SHAP values indicate that higher feature values increase the probability of re-election, while negative values decrease it. Education spending growth ( $\Delta\text{educ}$ ), political competition (enc\_gol), and health spending growth ( $\Delta\text{health}$ ) are the most important numeric predictors after region fixed effects. Dynasty also appears with a positive contribution, confirming that name recognition helps re-election.

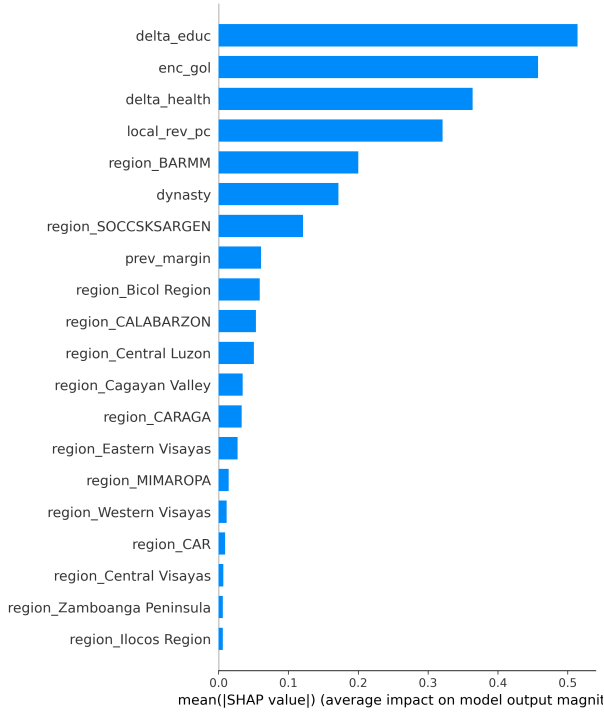
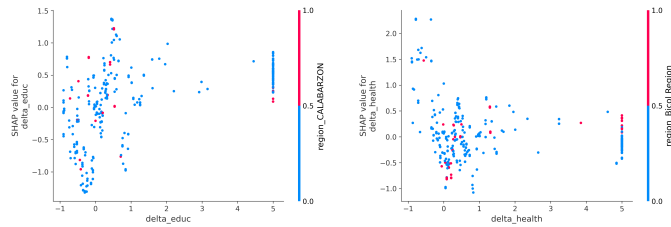


Fig. 3: SHAP bar plot (mean absolute SHAP values).

#### D. SHAP Dependence Analysis

Figures 4a and 4b show the SHAP dependence plots for education growth ( $\Delta_{educ}$ ) and health growth ( $\Delta_{health}$ ). The SHAP value increases with higher growth rates, indicating that voters reward governors who raise per-capita education and health spending. The effect is approximately linear for education and shows diminishing returns for health at very high growth rates.



(a) SHAP dependence for  $\Delta_{educ}$ . (b) SHAP dependence for  $\Delta_{health}$ .

Fig. 4: SHAP dependence plots for education and health spending growth.

#### E. Temporal Robustness

When trained on pre-2016 data and tested on 2019–2022, performance drops dramatically: accuracy 44.3%, AUC 0.412. This indicates that the model does not generalise to future elections.

#### F. Regression Discontinuity Results

Table I presents the RDD estimates for health spending growth. For the 5% bandwidth, winning a close election

increases health spending growth by 1.50 percentage points (95% CI [0.07, 2.94]). Figure 5 shows the discontinuity plot, and Figure 6 shows sensitivity to bandwidth.

TABLE II: RDD estimates (health spending growth)

| Bandwidth | ATE          | SE    | 95% CI                |
|-----------|--------------|-------|-----------------------|
| 3%        | 1.300        | 0.912 | [-0.487, 3.087]       |
| 5%        | <b>1.505</b> | 0.734 | <b>[0.066, 2.943]</b> |
| 7%        | 1.125        | 0.670 | [-0.187, 2.437]       |

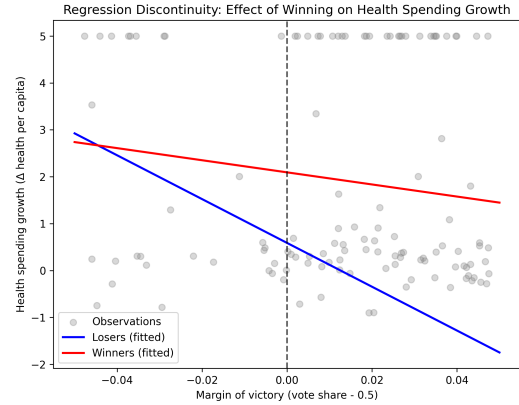


Fig. 5: Regression discontinuity plot (5% bandwidth).

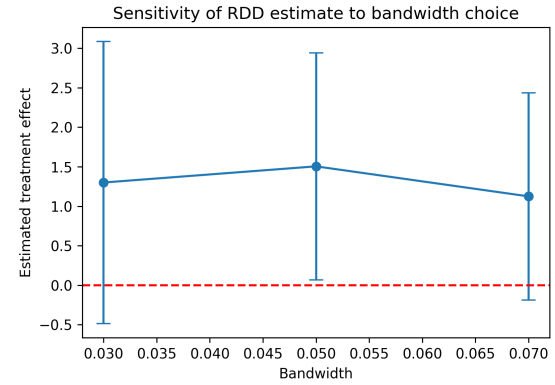


Fig. 6: Sensitivity of RDD estimate to bandwidth choice.

#### V. DISCUSSION

The random split results show strong correlation between fiscal performance and re-election. However, the temporal failure suggests that the relationship is not stable over time – possibly due to the COVID-19 pandemic or changing voter priorities. The RDD provides weak causal evidence: winning a close election increases health spending growth, but the effect is only significant at the 5% bandwidth.

Overall, the evidence supports the view that voters reward fiscal effort, but the effect is not robust across all specifications. Policy implications include: - **Invest in education and health:** Voters reward improvements in these sectors. - **Local**

**revenue effort:** Increasing local taxes is electorally beneficial (SDG 16). - **Dynasties have an advantage:** Anti-dynasty legislation may reduce incumbency advantage.

## VI. LIMITATIONS

- The predictive model is correlational, not causal.
- Temporal generalisation fails.
- The RDD effect is sensitive to bandwidth and only marginally significant.
- The dataset is limited to governors; mayors may behave differently.

## VII. CONCLUSION

This study demonstrates that fiscal performance correlates with gubernatorial re-election in the Philippines, but causal evidence is weak and temporal dynamics matter. The work contributes to SDG 16 (accountable institutions) and SDG 4 (quality education). Future research should incorporate time-varying effects and larger samples of close elections.

## ACKNOWLEDGMENT

This paper was originally presented at the 2nd International Conference on the UN SDGs and Social Innovation (ICUNSSI 2026) held on September 23–25, 2026, at the Mindanao State University-Iligan Institute of Technology, Iligan City, Philippines.

## REFERENCES

- [1] J. R. Lott Jr., “Attendance rates, political shirking, and the effect of post-elective office employment,” *Economic Inquiry*, vol. 28, no. 1, pp. 133–150, 1990.
- [2] L. S. Rothenberg and M. S. Sanders, “Legislative shirking and electoral accountability,” *Public Choice*, vol. 103, no. 3/4, pp. 313–335, 2000.
- [3] R. U. Mendoza, E. L. Beja Jr., V. S. Venida, and D. B. Yap, “Political dynasties and poverty: Measurement and evidence,” *Philippine Review of Economics*, vol. 49, no. 2, pp. 1–25, 2012.
- [4] J. J. Capuno, “Political dynasties and local fiscal performance: Evidence from the Philippines,” *Asian Journal of Political Science*, vol. 27, no. 2, pp. 123–145, 2019.
- [5] R. A. L. Pano, B. F. A. Mendoza, E. F. Habacon, and M. C. C. Reyes, “Philippine Local Government Interactive Dataset,” Center for Integrative and Development Studies, University of the Philippines. [Online]. Available: <https://elections.cids.up.edu.ph/>
- [6] D. S. Lee and T. Lemieux, “Regression discontinuity designs in economics,” *Journal of Economic Literature*, vol. 48, no. 2, pp. 281–355, 2010.
- [7] V. Chernozhukov et al., “Double/debiased machine learning for treatment and structural parameters,” *The Econometrics Journal*, vol. 21, no. 1, pp. C1–C68, 2018.
- [8] T. Chen and C. Guestrin, “XGBoost: A scalable tree boosting system,” in *Proc. 22nd ACM SIGKDD*, 2016, pp. 785–794.
- [9] S. M. Lundberg and S. I. Lee, “A unified approach to interpreting model predictions,” in *Advances in Neural Information Processing Systems*, 2017, pp. 4765–4774.